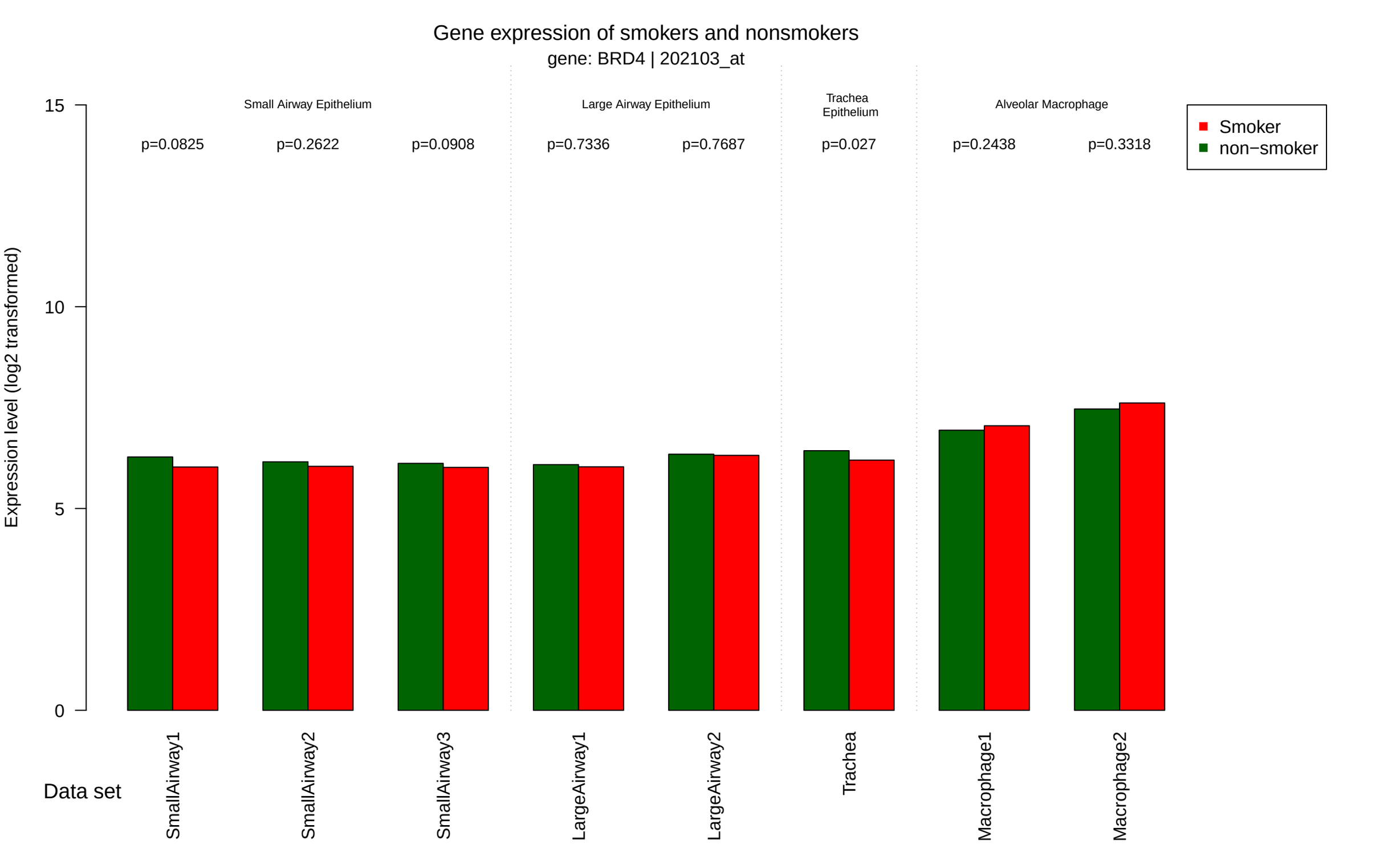
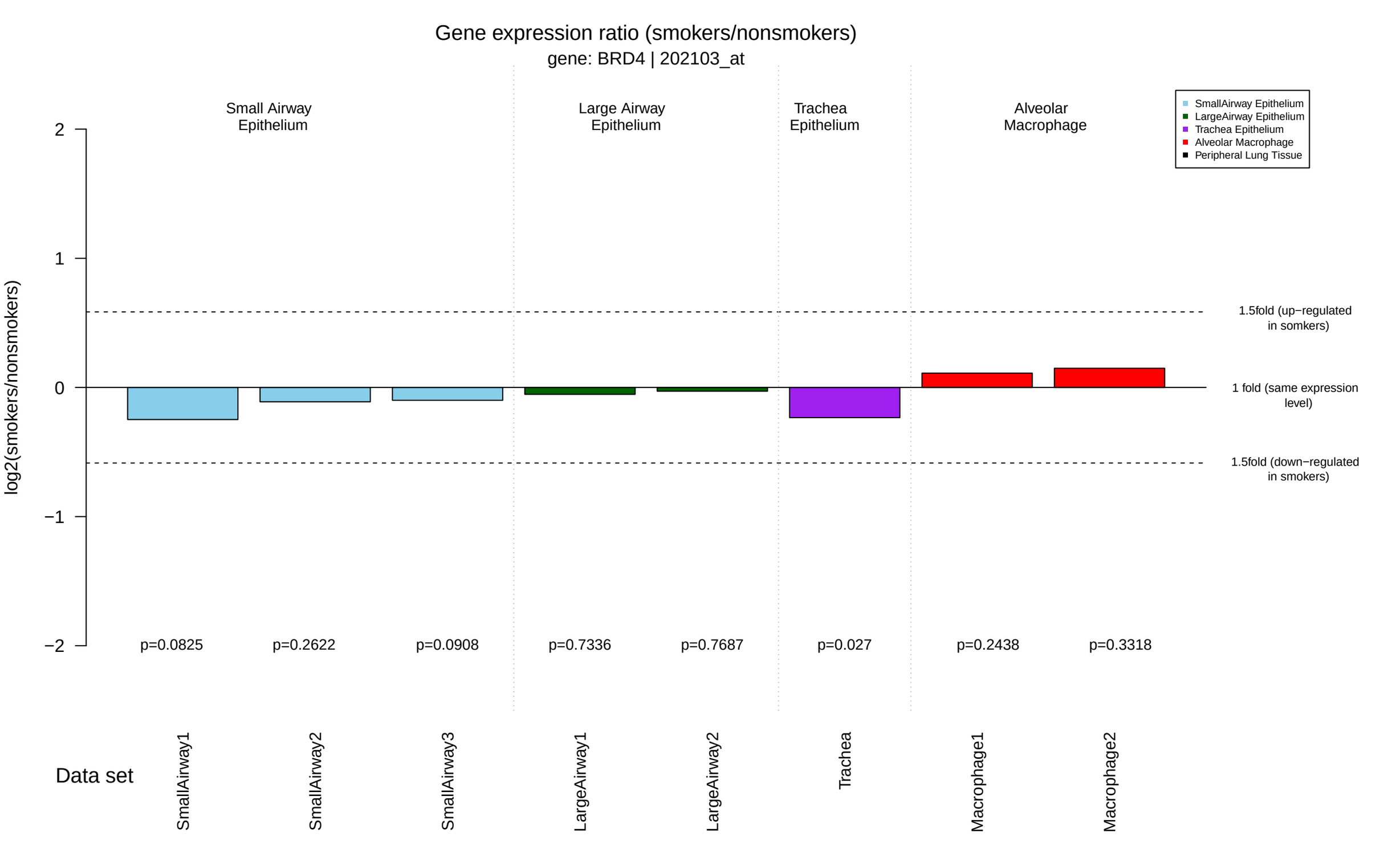
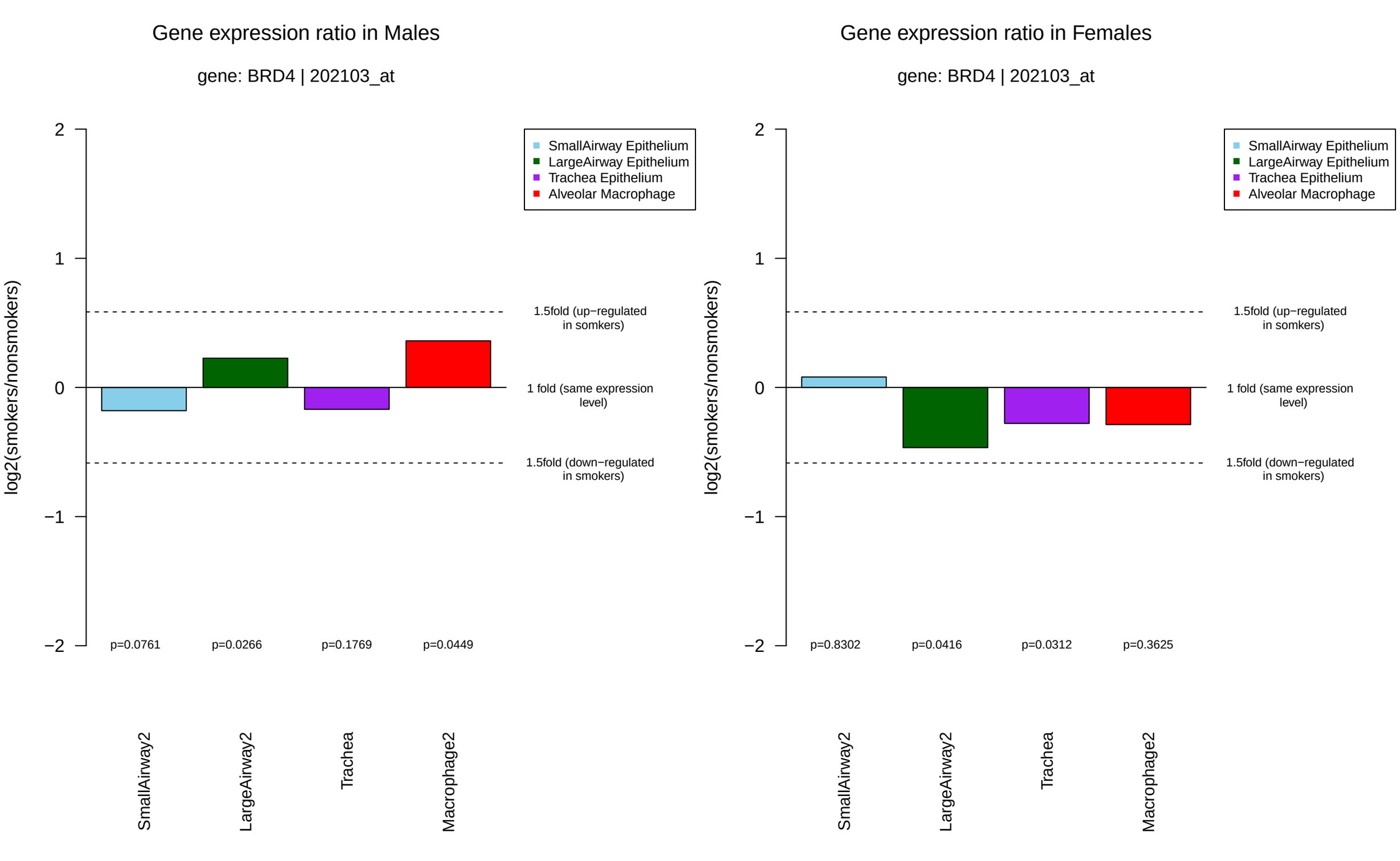
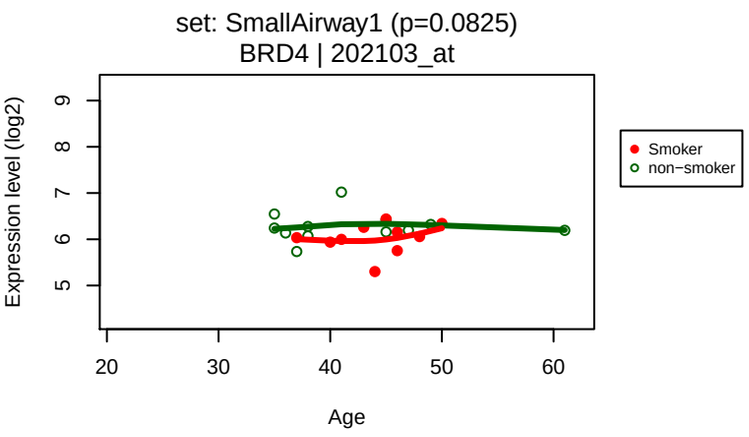
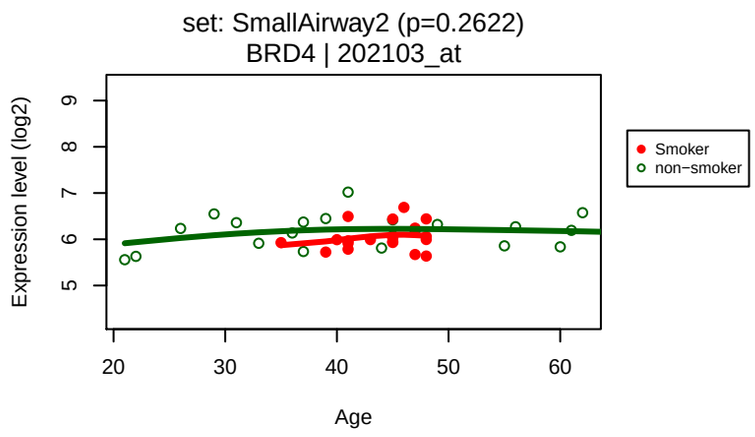
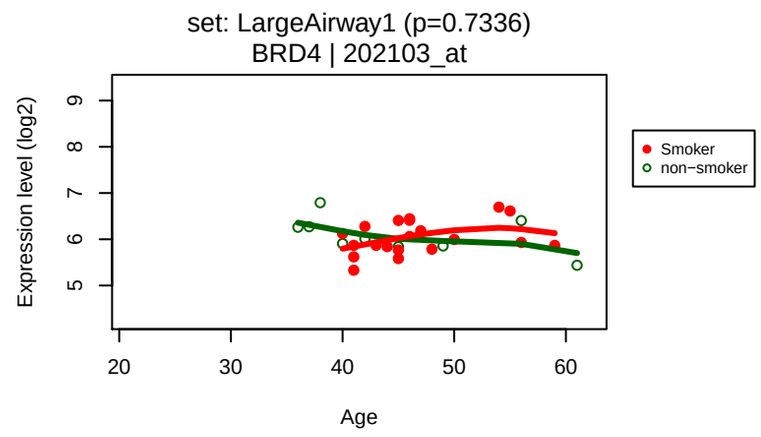

Gene expression of smokers and nonsmokers
gene: BRD4 | 202103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0825
p=0.2622
p=0.0908
p=0.7336
p=0.7687
p=0.027
p=0.2438
p=0.3318
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

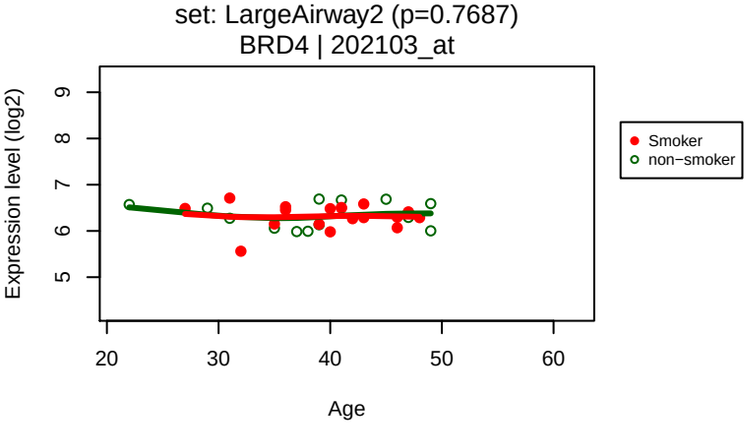

Gene expression ratio (smokers/nonsmokers)
gene: BRD4 | 202103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0825
p=0.2622
p=0.0908
p=0.7336
p=0.7687
p=0.027
p=0.2438
p=0.3318
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

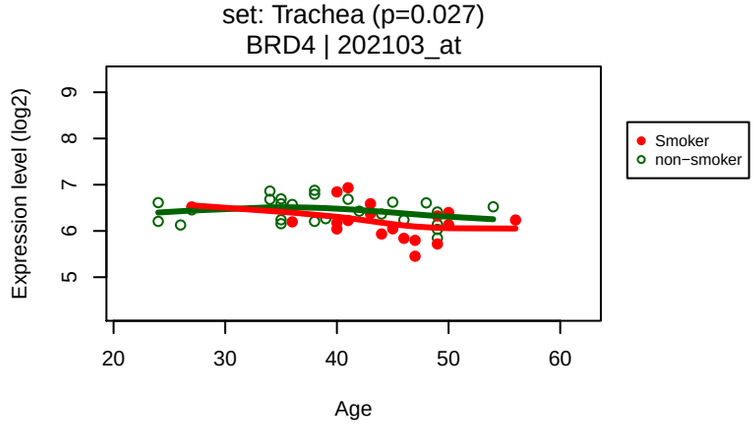

Gene expression ratio in Males
gene: BRD4 | 202103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0761
p=0.0266
p=0.1769
p=0.0449
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRD4 | 202103_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8302
p=0.0416
p=0.0312
p=0.3625
SmallAirway2
LargeAirway2
Trachea
Macrophage2

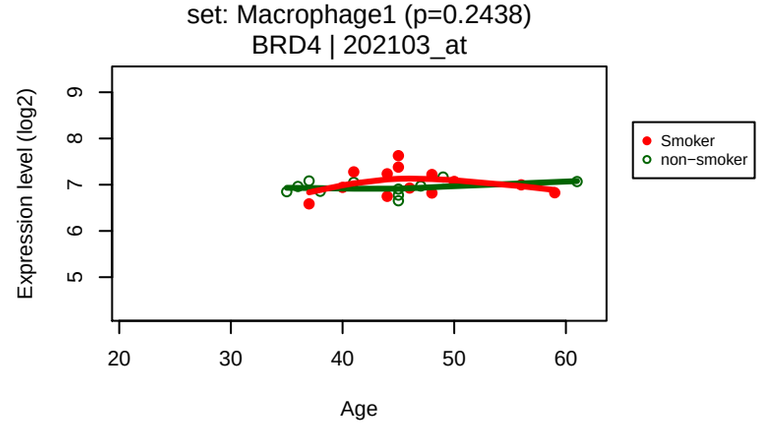

set: SmallAirway1 (p=0.0825)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker

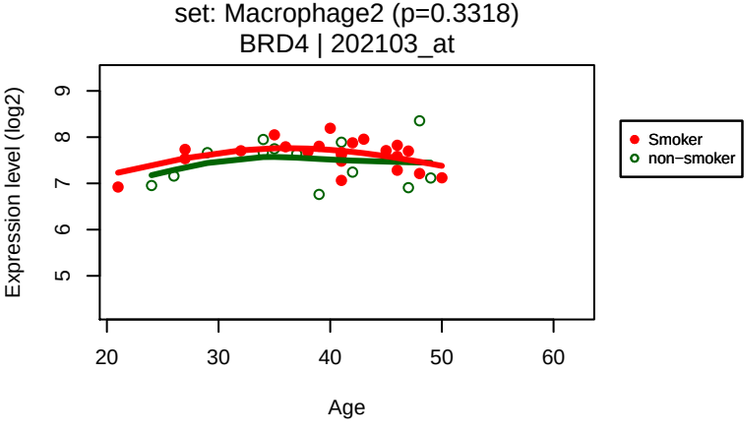

set: SmallAirway2 (p=0.2622)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker

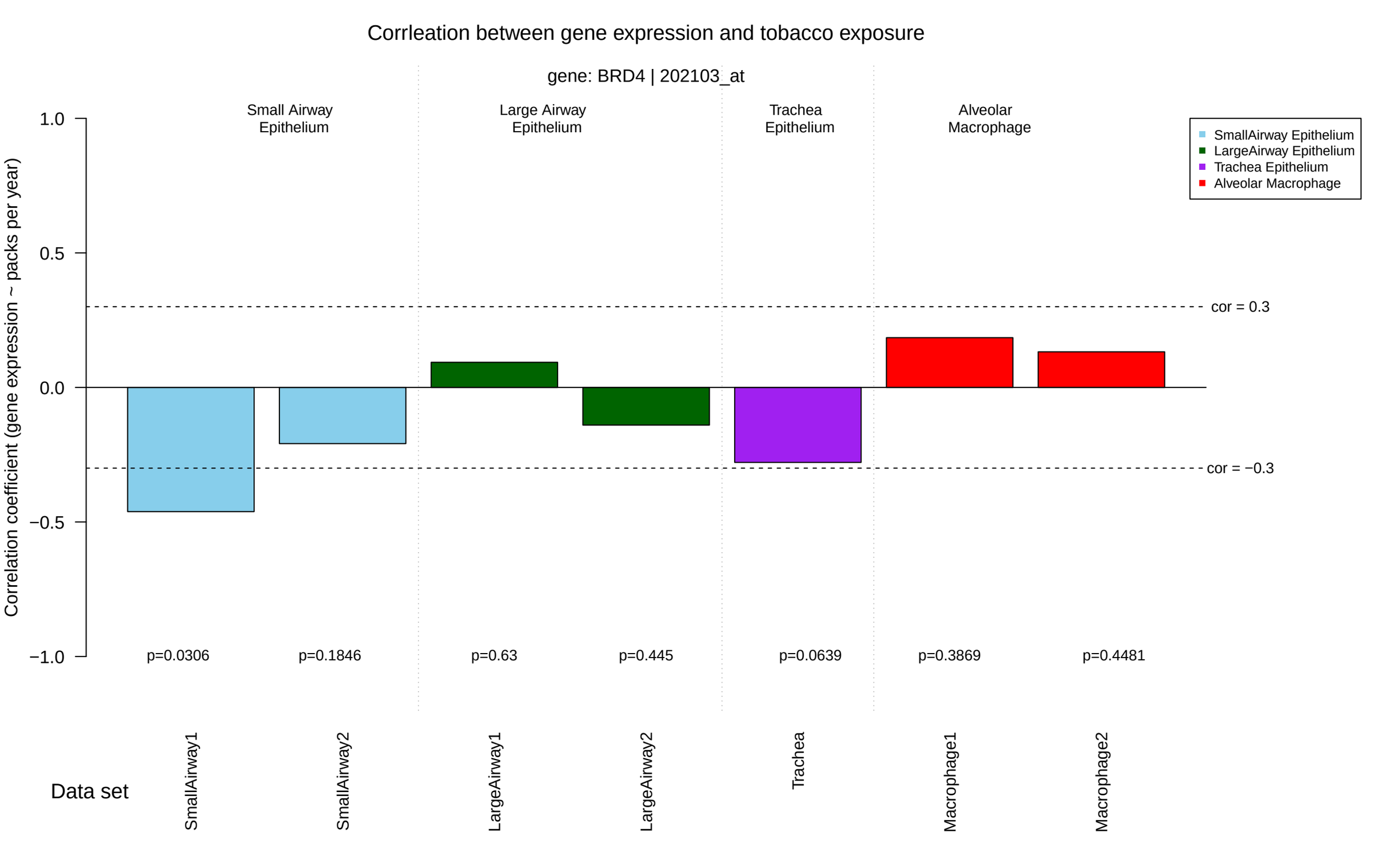

set: LargeAirway1 (p=0.7336)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7687)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.027)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2438)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3318)
BRD4 | 202103_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRD4 | 202103_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0306
p=0.1846
p=0.63
p=0.445
p=0.0639
p=0.3869
p=0.4481
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2